

B.Tech Degree VIII Semester (Supplementary) Examination

September 2010

ME 802 MACHINE DESIGN II (2002 Scheme)

(Design Data Book is permitted)
(Assume reasonable data wherever necessary)

Time: 3 Hours

Maximum Marks: 100

- I. (a) Explain any two type of friction clutches. (10)
(b) Design completely a flat belt drive to drive a winch from an electric motor of 8KW power. The speed of the motor shaft is 900 rpm. There is no considerable variation of load. Speed ratio is 4. (15)
- OR**
- II. (a) Explain band type brakes. (5)
(b) What are the different modes of chain failure? (5)
(c) Design a V belt drive to transmit 50 KW at 1440 rpm from an electric motor to a textile machine running 24 hours a day. The speed of the machine shaft is 480 rpm. (15)
- III. (a) Explain interference in involute profiles. (5)
(b) Describe nomenclature of a spurgear. (5)
(c) A motor shaft rotating at 1500 rpm has to transmit 15 KW to a low speed shaft with a speed reduction 3:1. The teeth are $14\frac{1}{2}^\circ$ involute with 25 teeth on pinion. Both the pinion and gear are made of steel with a maximum safe stress of 200 MPa. A safe stress of 40 MPa may be taken for the shaft on which the gear is mounted and for the key. Design a spur gear drive to suit the above conditions. (15)
- OR**
- IV. (a) Explain law of gearing. (5)
(b) Derive Buchingham equation for dynamic load. (10)
(c) A pair of Bevel gears with 20° pressure angle, consists of a 20 teeth pinion meshing with a 30 teeth gear. The module is 4mm while the face width is 20mm. The pinion rotates at 500 rpm and receives power from the electric motor. The starting torque of the motor is 150% of the rated torque. Determine the rated power that the gear can transmit. (10)
- V. (a) Explain desirable properties of lubricants. (5)
(b) Explain hydrodynamic theory of lubrication. (5)
(c) A 30 s work cycle consists of two parts

	Part I	Part II
Duration (s)	10	20
Radial load(KN)	45	15
Axial load (KN)	12.5	6.25
Speed (rpm)	720	1440

For this application the static and dynamic load capacities of a single-row deep groove ball bearing are 50 and 68 KN respectively. Calculate the life of the bearing in hours. (15)

OR

(P.T.O)



- VI. (a) Explain the nomenclature of a journal bearing. (10)
(b) Design a journal bearing for a centrifugal pump from the following data. Load on the journal is 20,000 N. Speed of journal is 900 rpm. Type of oil used is SAE 10. Ambient temperature of oil is 15.5°C . Maximum bearing pressure for the pump is 1.5N/mm^2 . Also calculate mass of the lubricating oil required for artificial cooling, if rise of temperature of oil be limited to 10°C . (15)
- VII. (a) Explain design recommendations for castings. (15)
(b) Describe design recommendations for screw threads. (10)
- OR**
- VIII. (a) Explain design considerations for forgings. (10)
(b) Describe the preparation of working drawings for manufacture of parts. (10)
(c) Explain different types of tolerance. (5)

ME (2000) - 6

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2003 - 1

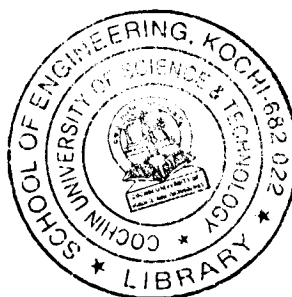
2002 - 3

B.Tech Degree VIII Semester Examination, April 2010**ME 803 REFRIGERATION AND AIR CONDITIONING***(2002 Scheme)*

Time: 3 Hours

Maximum Marks: 100

- I. (a) Explain with neat sketch a Bootstrap cycle of air refrigeration system. (10)
 (b) A cold storage plant is required to store 150 tons of fish. The temperature at which the fish was supplied is 35°C . Storage temperature of the fish is -10°C . C_p of the fish above freezing point is 2.49 KJ/KgK . C_p of the fish below freezing point is 1.26 KJ/KgK . Freezing point is -5°C . Latent heat of fish is 250 KJ/Kg . The actual COP is one third the Carnot COP. If the cooling is achieved within half a day, find:
 (i) Capacity of the plant (ii) Carnot COP (iii) Power required to run the plant. (10)
- OR**
- II. (a) Explain the working of a Bell-Coleman cycle and determine the expression for C.O.P. (10)
 (b) A refrigerator working on Bell-Coleman cycle (Reverse Brayton cycle) operates between 1 bar and 10 bar. Air is drawn from cold chamber at -10°C . Air coming out of compressor is cooled to 50°C before entering the expansion cylinder. Polytropic law $P.V^{1.3} = \text{constant}$ is followed during expansion and compression. Find theoretical C.O.P of the origin. Take $\gamma = 1.4$ and $C_p = 1.00 \text{ kJ/kg}^{\circ}\text{C}$ for air. (10)
- III. (a) Give comparison between vapour compression and absorption system. (10)
 (b) A simple vapour compression refrigeration system operating on R22 is working between the temperature limits of -10°C and 30°C . There is a subcooling of 10°C and superheating of 5°C . The mass flow rate is 0.5 Kg/s . Determine the cooling capacity and COP of the plant. (10)
- OR**
- IV. (a) Draw a neat diagram of Lithium bromide water absorption system and explain its working. (10)
 (b) Draw a neat diagram of 'Electrolux refrigerator' and explain its working principle. What is the important role of hydrogen in this system? (10)
- V. (a) What are the functions of a refrigeration compressor? List the types of Refrigerating compressors and give a brief explanation. (10)
 (b) Explain briefly the following types of reciprocating compressors:
 (i) Open type (ii) Semi sealed type (iii) Hermetically sealed type. (10)
- OR**
- VI. (a) Explain with neat sketches the following evaporators:
 (i) Flooded type (ii) Dry expansion type (10)
 (b) Explain the operation of a capillary tube in a refrigeration system. Why it is preferred to other throttling devices in household refrigerators? (10)
- VII. (a) 30 m^3 of air at 15°C DBT and 13°C WBT is mixed with 12 m^3 of air at 25°C DBT and 18°C WBT. Assuming the barometric pressure of one std atmosphere, determine the DBT and WBT of the resulting mixture. (10)
 (b) Explain summer air conditioning system and winter air conditioning system. (10)

OR*(Turn Over)*

- VIII. (a) Atmospheric air at 35°C and 60% RH is conditioned to 22°C DBT and 55% RH. This is done first by cooling and dehumidifying and then heating. If the air flow rate is 60m^3 per minute, find the following:
(i) Mass of water drained (ii) Capacity of cooling coil
(iii) Capacity of heating coil
(b) What is an effective temperature? Explain the factors which govern the optimum effective temperature.
- IX. (a) What is infiltration? Explain briefly how air leakage through window and door cracks can be estimated.
(b) An air conditioned room is maintained at 27°C (DBT) and 50% RH, while the outside conditions are 40°C DBT and 27°C WBT. The room sensible heat gain is 14 KW. The air enters the conditioned hall at 7°C DBT and saturated. Determine:
(i) Volume of moist air supplied to the room in m^3/min . (ii) Latent heat gain in room in KW (iii) Cooling load of the air washer in KW if 30% of the air supplied is fresh air and remaining 70% is recirculated.
- OR**
- X. (a) What is the function of ducts? Enumerate duct systems and explain any two of them with neat sketches.
(b) Enumerate the sources of noise in air-conditioning systems. Explain briefly the commonly used methods for noise control.

B.Tech Degree VIII Semester (Supplementary) Examination September 2010

ME 803 REFRIGERATION AND AIR CONDITIONING (2002 Scheme)

Time: 3 Hours

Maximum Marks: 100

- I. (a) Write notes on vortex tube and pulse tube refrigeration systems. (8)
 (b) The overall volume compression ratio of a reversed carnot cycle working with air as a refrigerant is 10. The temperature limits of the cycle are 40°C and 0°C. Determine –
 (i) the pressure, volume and temperature at each point of the cycle
 (ii) the work done in the cycle
 (iii) the refrigerating effect
 (iv) the COP of the cycle. (12)
- OR**
- II. (a) Write notes on steam jet refrigeration and vortex tube refrigeration. (8)
 (b) The following performance is expected in a certain Bootstrap system :
 Turbine efficiency $\eta_T = 85\%$
 Secondary compressor efficiency $\eta_c = 77\%$
 Secondary heat exchanger effectiveness, $\epsilon = 0.9$
 The cabin pressure is maintained at one standard atmosphere. The cooling air temperature entering the secondary heat exchanger is 32°C. The compressed air leaves the 1° heat exchanger at 64°C. The refrigerated air is required to enter the cabin at 4.5°C. Calculate –
 (i) the temperature of air entering the cooling turbine
 (ii) the pressures of air at discharge from primary to secondary compressors. (12)
- III. (a) In a vapour compression cycle saturated liquid refrigerant 22 leaving the condenser at 40°C is required to expand to the evaporator temperature of 0°C in a cold storage plant.
 (i) Determine the percentage saving in network of the cycle per Kg of the refrigerant if an isentropic expander could be used to expand the refrigerant in place of the throttling device.
 (ii) Also determine the percentage increase in refrigerating effect per Kg of refrigerant as a result of use of the expander. Assume that compression is isentropic from saturated vapour state at 0°C to the condenser pressure. (14)
 (b) Explain in brief which refrigerant/s would you choose for each of the following applications and why?
 (i) A cold storage of 100 TR capacity using reciprocating compressor.
 (ii) An 800 TR air conditioning plant using centrifugal compressor/s.
 (iii) A small capacity frozen food cabinet to maintain –30°C temperature. (6)
- OR**
- IV. (a) Refrigeration Engineers usually presume that if a R12 car air conditioner compressor is operated with R134a, it's cooling capacity would fall by about 10%. Examine this assumption by a realistic vapour compression cycle analysis. (6)
 (b) In a lithium bromide-water system the condenser and evaporator temperatures are 35 and 8°C respectively. The generator temperature is 85°C and the absorber temperature is 30°C. Assume a pressure drop of 2.5 mm Hg between the generator and condenser and 1 mm Hg between the evaporator and absorber. Determine the heat rejection rates in the condenser and absorber per unit of refrigerating capacity. Also find the heat input to the generator, and COP. (14)
- V. (a) The variation of volumetry efficiency with pressure ratio for a R134a compressor of an automotive car air conditioner is as follows.
- | | | | | | | |
|-----------|----|------|------|------|----|------|
| P_K/P_O | 2 | 3 | 4 | 5 | 6 | 7 |
| η_v | 80 | 79.5 | 72.5 | 66.5 | 61 | 56.7 |
- Somehow, the air flow over the condenser of the air conditioner gets completely blocked. Estimate the maximum pressure at discharge if the evaporation temperature is 5°C. (10)

- (b) Catalogue data of a water cooled condenser of a manufacturer gives the following details :
- Condensing temperature - 48.9°C
 - Water inlet temperature - 37.8°C
 - Water flow rate - 20.694 Kg/s
 - Capacity - 14.5 tons .

Estimate the capacity of this condenser with the same water flow rate but with an inlet temperature of 30°C and a condensation temperature of 42°C . The evaporation temperature may be assumed to be constant at 2.2°C .

(10)

OR

- VI. (a) Calculate the pressure of R134a as it expands through a capillary tube having a bore of 1.05 mm if $4.028 \times 10^{-2} \text{ Kg/s}$ of saturated liquid enters the capillary tube at a temperature of 45°C . Neglect heat transfer. Determine the temperature at which the condition of choked flow occurs as the expansion proceeds. (10)
- (b) A direct expansion cooling and dehumidifying coil for air has 1.27 cm OD and 1.12 ID serpentine copper tubing in each row. The fins are made of aluminium and are 1 mm thick. The spacing between the tubes is 1.27 cm and that between the fins corresponds to 5 fins per cm. The tube arrangement is staggered and the spacing between the rows is 1.27 cm . Determine - (i) The ratio of the total finned surface area to the inside tube surface area (ii) The total finned surface area per unit face area of the cooling coil (iii) Fin efficiency (iv) Overall heat transfer co-efficient accounting for the thermal resistance of the condensate layer, tube wall and the refrigerant film. The refrigerant side heat transfer co-efficient is $1100 \text{ W m}^{-2} \text{ K}^{-1}$. (10)

- VII. A summer air conditioning plant mixes 70 cmm of outside air at 35°C DB and 23°C WB with 210 cmm of return air at 24°C DB and $50\% \text{ RH}$. The mixture passes over a cooling coil. Air off the coil is $90\% \text{ RH}$. The room SHF is 0.7 . (i) Find the ADP, and air off the coil dew point and dry bulb temperatures (ii) How much cooling in KW is the unit doing (iii) How much of the total load is sensible, and how much is latent? (20)

OR

- VIII. A conditioned room with partial recirculation of room air is to be maintained at 24°C DB , 17°C WB . The local outside environment conditions are 35°C DB , 26°C WB . The sensible heat load is 48 KW . The latent heat load from occupants and infiltration, but excluding ventilation, is 57 KW . Based on the occupancy, 37 cmm of ventilation air is required. Find (i) the temperature of air entering the room (ii) the volume of air passing through the room (iii) the state of air entering the air conditioner (iv) the required ADP and (v) the system BPF. (20)

- IX. A laboratory having an unusually large latent heat gain is required to be air conditioned. The design conditions and loads are as follows :

Summer design conditions	: 40°C DBT , 27°C WBT
Inside design conditions	: 25°C DBT , $50\% \text{ RH}$
Room sensible heat	: 34.9 KW
Room latent heat	: 18.6 KW

The ventilation air requirement is 85 cmm . Determine the following :

- (i) Ventilation load (ii) Room and effective sensible heat factors (iii) Apparatus dew point and amount of reheat for economical design (iv) Supply air quantity (v) Condition of air entering and leaving coil and supply air temperature (vi) Grand total heat. Assume suitable bypass factor. (20)

OR

- X. An air conditioned space is maintained at 26°C DBT and $50\% \text{ RH}$ when the outside conditions are 35°C DBT 28°C WBT .
- (i) If the space has a sensible heat gain of 17.6 KW and air is supplied to the room at a condition of 8°C saturated, calculate - (i) the mass and volume flow rates of air supplied to the room (ii) the latent heat gain of the space (iii) the cooling load of the refrigeration plant if 15% of total weight of air supplied to the space is fresh air, and the remainder is re circulated air.
- (ii) If the supply air to the room is to be maintained at a level of at least 6 cmm per ton of the cooling load, find (i) the new supply of air temperature (ii) the mass and volume flow rates of the supply of air, and recirculated air added after the A/c apparatus. (20)

B.Tech Degree VIII Semester Examination, April 2010

ME 801 OPERATIONS MANAGEMENT
(2006 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART – A
(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Explain algorithm for finding critical path.
(b) Compare CPM and PERT.
(c) What is “Product Life Cycle”.
(d) Explain “Bin Card System”.
(e) Describe “Gantt Chart”.
(f) What are the functions of Despatch.
(g) What are the principles of material handing.
(h) What is “Depreciation”

PART – B
(All questions carry EQUAL marks)

(15 x 4 = 60)

- II. Draw the PERT network and calculate project completion time. The following informations are available.

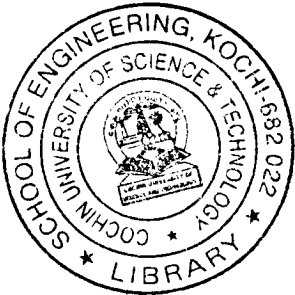
Predecessor activities	A	B	C	D	E	F	G	H	I	J	K	L
Successor activities	D F	L E	B H	G H	G K	I K	I K	L L	L H	G H	-	-
Duration (week)	8	9	10	7	6	9	4	3	3	12	10	16

OR

- III. The precedence relationship with Node numbers of a maintenance job are given as follows:
(i) Draw the arrow diagram and show the critical path (ii) indicate the slack times for jobs (3, 5) and (7, 8) (ie.) for ‘d’ and ‘l’.

Job	Node Number	Estimated Duration (Days)
a	(1, 2)	2
b	(2, 3)	3
c	(2, 4)	4
d	(3, 5)	5
e	(3, 6)	10
f	(4, 6)	6
g	(4, 7)	3
h	(5, 8)	8
k	(6, 8)	7
l	(7, 8)	3

(Turn Over)



- IV. Write notes on: (i) Interchangeability (ii) Standardisation
(iii) Product design

OR

- V. A concern is engaged in casting of carburetors. It's demand from the automobile industries per year is 50,000. The cost of set up is Rs.1000/-. There are ten workers engaged, each on wage rate of Rs. 15/- per day. The overhead cost is Rs.100/- per day. The daily production capacity is 200/-. The material cost for each carburetor is Rs.10/-. The annual rate of depreciation, insurance, taxes and storage cost, etc. is 20% of unit cost. The supply should be instantaneous and no shortages are permitted.

Find: (i) The economic lot size (ii) The number of runs
(iii) Duration of each run

- VI. Determine an optimum sequence to process the various types of fan blades each day from the following information so as to minimize the total elapsed time.

Type of fan Blade	Number to be Processed in each day	<u>Processing time on</u>	
		Machine A (Minutes)	Machine B (Minutes)
1	4	4	8
2	6	12	6
3	5	14	16
4	2	20	22
5	4	8	10
6	3	18	2

Also workout the total elapsed time for an optimum square. What is the total idle time on Machine A and on Machine B.

OR

- VII. We have five jobs each of which must go through three machines A, B and C in the order A, B, C. Processing time in hours are given. Determine a sequence for the five jobs that will minimize the total elapsed time. Find also the idle time for machine A, B and C.

Job	<u>Processing Time</u>		
	Ai	Bi	Ci
1	16	10	8
2	20	12	18
3	12	4	16
4	14	6	12
5	22	8	10

- VIII. (a) Explain the factors to be considered for plant location.
(b) What are the different types of plant layouts. Explain any two in detail.

OR

- IX. (a) Differentiate between preventive maintenance and breakdown maintenance.
(b) How can we find the optimum period for replacement? Show graphical representation.

ME 2006

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B. Tech Degree VIII Semester (Supplementary) Examination September 2010

ME 801 OPERATIONS MANAGEMENT (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART - A (All questions carry EQUAL marks)

(8 x 5 = 40)

- I. (a) Define following terms – critical activities, non-critical activities and dummy activities.
 (b) Write short notes on Time Series methods of forecasting.
 (c) Explain the functions of Production Planning and Control.
 (d) Derive an expression for Economic Order Quantity when stock replenishment is not instantaneous.
 (e) What is Gantt Chart? How it is constructed?
 (f) What is dispatching? What are the functions of dispatching list?
 (g) Compare product layout and process layout.
 (h) Describe the objectives of material handling.

PART - B

(4 x 15 = 60)

- II. (a) Bring out the difference between CPM and PERT.
 (b) A small project is composed of the following activities whose time estimates are given below. Time is indicated in days.

ACTIVITY	A	B	C	D	E	F	G	H	I
OPTIMISTIC TIME	3	2	3	2	2	3	2	5	3
MOST LIKELY TIME	3	4	4	3	5	6	5	8	6
PESSIMISTIC TIME	8	8	10	4	14	15	8	11	15
PRECEEDING ACTIVITY	—	—	—	A	B	C	D,E	D,E	F,G

- (i) Construct Network and find expected duration and variance of each activity.
 (ii) What is critical path?
 (iii) If the project due date is 28 days, what is the probability of not meeting due date?
 (iv) Calculate the project due date for probability of completion of 85%. (12)

OR

- III (a) Write short notes on Judgmental methods for forecasting.
 (b) ABC Ltd. has started the production of invertors. The demand for different weeks of March and April of the company are given below. Assume the initial forecast for the first week of March as 6000 and corresponding initial trend as zero. Assume $\alpha = 0.1$ and $\beta = 0.2$. Represent the trend adjusted exponential smoothing forecast for March and April. Also find MAD, MSE, RSFE, MAPE and tracking signal. (3)

(P.T.O)

Month	March				April			
Week	1	2	3	4	1	2	3	4
Demand	6500	6000	5500	6500	6500	6250	7000	7100

(12)

- IV. (a) Explain the various stages of product life-cycle. (7)
 (b) Write notes on Simplification, Standardization, Specialization and Interchangeability. (8)

OR

- V. (a) Write short notes on :
 (i) Re-order point
 (ii) Lead time
 (iii) Safety stock
 (iv) Average inventory (5)
 (b) Annual demand for an item is 4800 units. Ordering cost is Rs.500/- per order. Inventory carrying cost is 18% of the purchase cost per unit per year. The price brakes as given below. Find Optimum order quantity. If the ordering cost is changed to Rs.350/-, what will be optimum ordering quantity?

Quantity	Unit Price
$0 \leq Q < 1200$	10
$1200 \leq Q < 2000$	9
$2000 \leq Q$	8

(10)

- VI. (a) What do you mean by sequencing? Explain algorithm for solving two job n machine problem. (9)
 (b) Write notes on progress reporting and expediting. (6)

OR

- VII. (a) Explain the advantages and disadvantages of various aggregate planning methods. (7)
 (b) The following parts are waiting to be processed in a machine shop on July 23. The estimate of time needed to complete the part is as follows.

Parts	1	2	3	4	5
Due-date	July 31	August 2	August 16	July 29	August 30
Processing time	10	7	24	6	30

Sequence the parts using critical ratio method.

(8)

- VIII. (a) What is the objective of maintenance? Explain in detail the different types of Maintenance Procedures. (8)
 (b) What are the principles of material handling? (7)
- OR
- IX. (a) What are the different factors influencing plant location? What are location factors? (5)
 (b) How will you classify the production systems? Explain. (5)
 (c) What is the significance of group layout? (5)

B. Tech Degree VIII Semester (Supplementary) Examination September 2010

ME 802 COMPRESSIBLE FLUID FLOW (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

*(Use of approved gas tables permitted)***PART - A**(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Define stagnation temperature and stagnation pressure.
 (b) Explain various regions of flow with the help of steady flow ellipse.
 (c) Explain the development of a normal shock wave. Explain whether a rare fraction shock can be developed in a flow field.
 (d) Write the equation relating density ratio and pressure ratio across a shock wave. What is the corresponding equation in the case of isentropic flow?
 (e) Draw fanno curve on $h-s$ diagram and mark all important points on it. List the assumptions followed.
 (f) Represent Rayleigh line on $h-s$ plot and prove that the limit of heating and cooling processes correspond to $M = 1$.
 (g) Explain shock tube.
 (h) Explain temperature recovery factor.

PART - B

(4 x 15 = 60)

- II. (a) Derive the relation for stagnation temperature $\left(\frac{T_0}{T}\right)$ in terms of local Mach number M . (5)
 (b) The pressure, temperature and Mach number at the entry of a flow passage are 2.45 bar, 26.5°C and 1.4 respectively. If the exit Mach number is 2.5 determine for adiabatic flow of a perfect gas ($\gamma = 1.3$, $R = 0.469 \text{ KJ/Kg-K}$)
 (i) stagnation temperature (ii) temperature and velocity of gas at exit
 (iii) the flow rate per square metre of the inlet cross section. (10)

OR

- III. (a) Explain choking in nozzles. (5)
 (b) A conical diffuser has entry and exit diameters of 15 cm and 30 cm respectively. The pressure, temperature and velocity of air at entry are 0.69 bar, 340 K and 180 M/s respectively. Determine (i) exit pressure (ii) the exit velocity (iii) force exerted on diffuser walls. (Assume isentropic flow, $\gamma = 1.4$ $C_p = 1.00 \text{ KJ/Kg-K}$). (10)

(P.T.O.)

- IV. (a) Derive Rankine Hugoniot relation for a normal shock wave. (8)
 (b) A gas ($\gamma = 1.4$, $R = 0.287 \text{ KJ/Kg-K}$) at a Mach number of 1.8, $P = 0.8 \text{ bar}$ and $T = 373 \text{ K}$ passes through a normal shock. Determine its density after the shock. (7)
- OR**
- V. (a) Explain oblique shock waves with examples. (5)
 (b) A normal shock occurs in the diverging section of a convergent – divergent air nozzle. The throat area is $1/3^{\text{rd}}$ exit area and the static pressure at exit is 0.4 times the stagnation pressure at the entry. The flow is isentropic throughout except through the shock. Determine (i) the Mach numbers M_x and M_y (ii) the static pressure (iii) the area of cross section of the nozzle at the section of the nozzle where the normal shock occurs. (10)
- VI. (a) Starting from the relation $\frac{dT}{T} = - \frac{\gamma - 1}{2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)} dM^2$ obtain the variation of temperature between two sections 1 and 2. (5)
 (b) Air at $P_0 = 10 \text{ bar}$, $T_0 = 400 \text{ K}$ is supplied to a 50 mm dia pipe. The friction factor of the pipe surface is 0.002. If the Mach number changes from 3.0 at the entry to 1.0 at the exit determine (i) length of the pipe (ii) the mass flow rate. (10)
- OR**
- VII. (a) For Rayleigh flow, starting from fundamentals obtain the relation for the pressure ratio between two sections in terms of the mach numbers there. (5)
 (b) The conditions of a gas in a combustor at entry are $P_1 = 0.343 \text{ bar}$, $T_1 = 310 \text{ K}$, $C_1 = 60 \text{ m/s}$. Determine the mach number, pressure, temperature and velocity at the exit if the increase in stagnation enthalpy of the gas between entry and exit is 1172.5 KJ/Kg . Take $C_p = 1.005 \text{ KJ/Kg-K}$, $\gamma = 1.4$. (10)
- VIII. Explain constant temperature and constant current type hot wire anemometer. (15)
- OR**
- IX. Explain both open circuit and closed circuit supersonic wind tunnel. (15)

B. Tech Degree VIII Semester Examination, April 2010**ME 802 COMPRESSIBLE FLUID FLOW***(2006 Scheme)*

Time : 3 Hours

Maximum Marks : 100

PART – A(Answer ALL questions)(All questions carry EQUAL marks)

(8 x 5 = 40)

(Use of steam tables and gas tables are permitted)

- I. (a) What is meant by steady flow ellipse? Indicate various regions of flow on this ellipse.
 (b) Explain choking in isentropic flow.
 (c) Write down the Prandtl's relation for stationary normal shock and hence show that for a given stagnation temperature the following relation is valid.

$$M_1^2 M_2^2 = \text{Const} / T_1 T_2$$

- (d) Sketch the Mach wave pattern that occurs when a supersonic flow is turned away from itself with the help of a convex corner.
 (e) Explain how the friction in a constant area duct affects the flow, if the flow is
 (i) initially subsonic (ii) initially supersonic
 (f) What is Rayleigh flow? Under what condition does the assumption of Rayleigh flow is invalid for a heat exchanger?
 (g) How is it possible to employ the pitot tube for measurement in a supersonic stream? Derive the Rayleigh supersonic pitot formula.
 (h) How does a shadowgraph preferred over a Schlieren system in studying density effects?

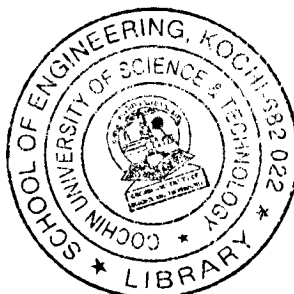
PART – B

(4 x 15 = 60)

- II. (a) The pressure temperature and Mach number at the entry of a flow passage are 2.45 bar, 26.5°C and 1.4 respectively. If the exit Mach number is 2.5 determine for adiabatic flow of a perfect gas ($\gamma = 1.3$, $R = 0.469 \text{ kJ/KgK}$):
 (i) Stagnation temperature
 (ii) Temperature and velocity of gas at exit
 (iii) The flow rate per square metre of the inlet cross section. (8)
 (b) Derive the general equation of conservation of energy as applied to a control volume. Hence show that for one-dimensional study flow of a perfect gas, the energy equation can be written as $h + \frac{C^2}{2} = \text{constant}$. (7)

OR

- III. (a) Air enters a variable area duct at a pressure of 350 kN/m², a temperature of 27° and a velocity of 150 m/s. The area of cross-section at entry is 500 cm² and the exit area is 350 cm². If the flow is isentropic, determine
 (i) the mass flow rate through the duct
 (ii) the pressure, temperature and velocity at exit from the duct
 (iii) the change in impulse function. (9)
 (b) Derive an expression for area ratio A/A^* in terms of Mach number, for an isentropic flow of a perfect gas. (6)

(Turn Over)

- IV. A certain nozzle is designed for an isentropic flow with an exit Mach number 2. Find the pressure ratio which locates a normal shock at
- the diverging throat of the nozzle
 - exit of the nozzle
 - half way down the diverging section of the nozzle.
- Assume $\gamma = 1.4$ and that the diverging section is having uniform divergence. (15)
- OR**
- V. (a) Starting from the equation for entropy change across a normal shock wave, find an equation for entropy change across an oblique shock wave. (5)
- (b) Air having a Mach number 3.0, approaches a symmetrical wedge having a wedge angle of 30° . The pressure and temperature of the air are 1 bar and 27°C . Find the Mach number and velocity of flow downstream of the shock wave, assuming that a weak oblique shock is formed. Also find the pressure, density, temperature and stagnation pressure downstream of the shock wave. (10)
- VI. (a) Show that the Mach number is unity at the point of maximum entropy on the Fanno line. (7)
- (b) Air flows through an insulated circular pipe at a rate of 495 Kg/minute. The pressure, temperature and Mach number of air at entrance to the pipe are 0.3MPa, 27°C and 0.15 respectively. The co-efficient of friction for the pipe is assumed constant and its value is 0.005. If the Mach number at exit is 0.5, determine
- the diameter of the pipe
 - the length of the pipe
 - pressure and temperature of air at the exit of the pipe
 - stagnation pressure loss. (8)
- OR**
- VII. A combustion chamber in a gas-turbine plant receives air at 300K, 55 KPa and 60m/s. The fuel air ratio is 29 and the calorific value of the fuel is 42MJ/Kg. Assuming $\gamma = 1.4$ and $R = 0.287\text{KJ/Kg.K}$ for the gas, determine
- Mach numbers at inlet and exit
 - Pressure, temperature and velocity of the gas at exit of Combustion chamber.
 - Percentage loss in stagnation pressure
 - Maximum attainable stagnation temperature. (15)
- VIII. (a) Explain with the aid of a schematic diagram the working of a closed circuit continuous supersonic wind tunnel. (7)
- (b) What are the two different arrangements by which the hot wire anemometer is used? With the help of circuit diagram and typical calibration curves explain how velocity can be calculated from the results. (8)
- OR**
- IX. (a) Explain how laser beam is employed in the measurement of velocity components in compressible flow? What are the advantages and disadvantages of laser velocimetry? (9)
- (b) Explain the working principle of shock tube with neat sketch. (6)

B. Tech Degree VIII Semester (Supplementary) Examination September 2010

ME 802 COMPRESSIBLE FLUID FLOW (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Use of approved gas tables permitted)

PART - A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Define stagnation temperature and stagnation pressure.
 (b) Explain various regions of flow with the help of steady flow ellipse.
 (c) Explain the development of a normal shock wave. Explain whether a rare fraction shock can be developed in a flow field.
 (d) Write the equation relating density ratio and pressure ratio across a shock wave. What is the corresponding equation in the case of isentropic flow?
 (e) Draw fanno curve on $h-s$ diagram and mark all important points on it. List the assumptions followed.
 (f) Represent Rayleigh line on $h-s$ plot and prove that the limit of heating and cooling processes correspond to $M = 1$.
 (g) Explain shock tube.
 (h) Explain temperature recovery factor.

PART - B

(4 x 15 = 60)

- II. (a) Derive the relation for stagnation temperature $\left(\frac{T_0}{T}\right)$ in terms of local Mach number M . (5)
 (b) The pressure, temperature and Mach number at the entry of a flow passage are 2.45 bar 26.5°C and 1.4 respectively. If the exit Mach number is 2.5 determine for adiabatic flow of a perfect gas ($\gamma = 1.3$, $R = 0.469 \text{ KJ/Kg-K}$)
 (i) stagnation temperature (ii) temperature and velocity of gas at exit
 (iii) the flow rate per square metre of the inlet cross section. (10)

OR

- III. (a) Explain choking in nozzles. (5)
 (b) A conical diffuser has entry and exit diameters of 15 cm and 30 cm respectively. The pressure temperature and velocity of air at entry are 0.69 bar, 340 K and 180 M/s respectively. Determine (i) exit pressure (ii) the exit velocity (iii) force exerted on diffuser walls. (Assume isentropic flow, $\gamma = 1.4$ $C_p = 1.00 \text{ KJ/Kg-K}$). (10)

(P.T.O.)

- IV. (a) Derive Rankine Hugoniot relation for a normal shock wave. (8)
 (b) A gas ($\gamma = 1.4$, $R = 0.287 \text{ KJ/Kg-K}$) at a Mach number of 1.8, $P = 0.8 \text{ bar}$ and $T = 373 \text{ K}$ passes through a normal shock. Determine its density after the shock. (7)
- OR**
- V. (a) Explain oblique shock waves with examples. (5)
 (b) A normal shock occurs in the diverging section of a convergent – divergent air nozzle. The throat area is $1/3^{\text{rd}}$ exit area and the static pressure at exit is 0.4 times the stagnation pressure at the entry. The flow is isentropic throughout except through the shock. Determine (i) the Mach numbers M_x and M_y (ii) the static pressure (iii) the area of cross section of the nozzle at the section of the nozzle where the normal shock occurs. (10)
- VI. (a) Starting from the relation $\frac{dT}{T} = - \frac{\gamma - 1}{2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)} dM^2$ obtain the variation of temperature between two sections 1 and 2. (5)
 (b) Air at $P_0 = 10 \text{ bar}$, $T_0 = 400 \text{ K}$ is supplied to a 50 mm dia pipe. The friction factor of the pipe surface is 0.002. If the Mach number changes from 3.0 at the entry to 1.0 at the exit determine (i) length of the pipe (ii) the mass flow rate. (10)
- OR**
- VII. (a) For Rayleigh flow, starting from fundamentals obtain the relation for the pressure ratio between two sections in terms of the mach numbers there. (5)
 (b) The conditions of a gas in a combustor at entry are $P_1 = 0.343 \text{ bar}$, $T_1 = 310 \text{ K}$, $C_1 = 60 \text{ m/s}$. Determine the mach number, pressure, temperature and velocity at the exit if the increase in stagnation enthalpy of the gas between entry and exit is 1172.5 KJ/Kg . Take $C_p = 1.005 \text{ KJ/Kg-K}$, $\gamma = 1.4$. (10)
- VIII. Explain constant temperature and constant current type hot wire anemometer. (15)
- OR**
- IX. Explain both open circuit and closed circuit supersonic wind tunnel. (15)

B. Tech Degree VIII Semester (Supplementary) Examination

September 2010

ME 804 (C) OPERATIONS RESEARCH (2002 Scheme)

Time : 3 Hours

Maximum Marks : 100

(All questions carry EQUAL marks)

- I. (a) Explain unit vectors, linear dependence and Euclidean spaces.
 (b) Test for consistency and solve
 $5x + 3y + 7z = 4,$
 $3x + 26y + 2z = 9,$
 $7x + 2y + 10z = 5.$

OR

- II. (a) Explain Hyper planes, Convex sets and Simultaneous equations.
 (b) Evaluate $3A - 4B$ where,

$$A = \begin{bmatrix} 3 & -4 & 6 \\ 5 & 1 & 7 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 0 & 3 \end{bmatrix}.$$

- III. (a) Explain Mathematical formulation of LPP.
 (b) Find the dual problem of the following .

$$\begin{aligned} \text{Minimise} \quad & Z = 5x_1 - 6x_2 + 4x_3 \\ \text{Subjected to} \quad & 3x_1 + 4x_2 + 6x_3 \geq 9 \\ & x_1 + 3x_2 + 2x_3 \geq 5 \\ & 7x_1 - 2x_2 - x_3 \leq 10 \\ & x_1 - 2x_2 + 4x_3 \geq 4 \\ & 2x_1 + 5x_2 - 3x_3 = 3 \\ & x_1, x_2, x_3 \geq 0 \end{aligned}$$

- (c) Explain Charnes 'M' Method.

OR

- IV. (a) Explain Graphical Method to solve LPP.
 (b) Tellitell Television Company operates two assembly lines, Line I and Line II newly commissioned. Each line is used to assemble the components of three types of televisions; CRT Type, CRT Slim Type and LCD Type. The expected daily production on each line is as follows.

TV Model		Line I (Old)	Line II (New)
CRT Type	—	3	1
CRT Slim Type	—	1	1
LCD Type	—	2	6

The daily running cost for two lines average Rs.6000/- for Line I and Rs.4000/- for Line II. It is given that the company must produce atleast 24 CRT Type, 16 CRT Slim Type and 48 LCD Type TV sets for which an order is pending
 You are required to formulate the above situation as LPP taking the objective functions as minimization of total cost. Also determine the number of days that the two should be run to meet requirements.

(P.T.O)

- V. A product is produced by four factories A,B,C and D. The unit production cost in them are Rs.2, Rs.3, Rs.1 and Rs.5 respectively. Their production capacities are; factory A = 50 units, B = 70 units, C = 30 units and D = 50 units. These factories supply the product to four stores, demands of which are 25,35,105 and 20 units respectively. Unit Transport Cost in rupees from each factory to each store is given in the table.

					<u>Stores</u>
	1	2	3	4	
	A	2	4	6	11
	B	10	8	7	5
	C	13	3	9	12
	D	4	6	8	3

Determine the extent of deliveries from each of the factory to each of the stores so that the total production and transportation cost is minimum.

OR

- VI. An air line that operates 7 days a week has the time table shown below. Crew must have a minimum lay over of 5 hours between flights. Obtain the pairing of flights that minimizes lay over time away from home assuming that the crew can be based at either of the two cities. The crew will be based at the city that results in smaller lay over.

Delhi – Jaipur

Flight No.	Depart	Arrive
201	7 AM	8 AM
202	8 AM	9 AM
203	1.30 PM	2.30 PM
204	6.30 PM	7.30 PM

Jaipur – Delhi

Flight No.	Depart	Arrive
101	8 AM	9.15 AM
102	8.30 AM	9.45 AM
103	12 Noon	1.15 PM
104	5.30 PM	6.45 PM

- VII. (a) What do you understand by zero sum and non zero-sum games?
 (b) Solve the following $2 \times n$ game by the method of sub games.

Player B

		B ₁	B ₂	B ₃
Player A	A ₁	1	3	11
	A ₂	8	5	2

OR

- VIII. (a) Explain pure and mixed strategies.

B.Tech Degree VIII Semester Examination, April 2010**ME 804 (C) OPERATIONS RESEARCH**
(2002 Scheme)(All questions carry EQUAL marks)

Time: 3 Hours

Maximum Marks: 100

- I. (a) Write notes on convex sets and their properties.
(b) Find the rank of the matrix.

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$$

OR

- II. Check for consistency and solve the following set of equations:
 $3X + 2Y + 2Z = 1$
 $X + 2Y = 4$
 $10Y + 3Z = -2$
 $2X - 3Y - Z = 5$

- III. Solve the following LP problem using simplex method.
 Maximize $Z = 10X_1 + 15X_2 + 20X_3$
 Subject to the constraints
 $2X_1 + 4X_2 + 6X_3 \leq 24$
 $3X_1 + 9X_2 + 6X_3 \leq 30$
 $X_1, X_2 \text{ and } X_3 \leq 0$

OR

- IV. Solve the following LP model using Big M method.
 Minimize $Z = 2X_1 + 3X_2$
 Subject to
 $X_1 + X_2 \geq 6$
 $7X_1 + X_2 \geq 14$
 $X_1, X_2 \geq 0$

- V. For the following transportation problem obtain the initial basic feasible solution using the north-west corner method and then optimize the solution using U-V method.

		Destination				Supply
		1	2	3	4	
Source	1	3	1	7	4	300
	2	2	6	5	9	400
	3	8	3	3	2	500
Demand		250	350	400	200	

The cell entries represent the cost of transportation per unit.

OR

(Turn Over)



- VI. Solve the following assignment problem.

	A	B	C	D
1	10	25	15	20
2	15	30	5	15
3	35	20	12	24
4	17	25	24	20

- VII. Solve the following game graphically.

	B_1	B_2	B_3
A_1	2	-3	-4
A_2	-6	-1	1

OR

- VIII. Solve the following game.

	B_1	B_2	B_3	B_4
A_1	3	2	4	0
A_2	3	4	2	4
A_3	4	2	4	0
A_4	0	4	0	8

- IX. A T.V. repairman finds that the time spent on his jobs has an exponential distribution with mean 30 minutes. If he repairs sets in the order in which they came in, and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hour day, what is repairman's expected idle time each day? How many jobs are ahead of the average set just brought in?

OR

- X. (a) Discuss in detail the application areas of queueing theory.
 (b) Explain Jockeying, Balking and Reneging with respect to a queueing system.